

MCQ Test

- 1) A) The regression line always passes through the mean of X and Y .
- 2) c) Decision trees can be used for classification and regression
- 3) c) Pre-pruning, pre Pruning
- 4) a) prone of overfitting
- 5) b) updating probabilities based on new evidence
- 6) b) The Probabilities of event A given event B has occurred.
- 7) d) 0.75
- 8) c) Total number of possible outcomes
- 9) a) 0.16
- 10) b) conditional Probability
- 11) a) The Probability of HHH increases
- 12) b) 4/11
- 13) a) Machine Learning
- 14) b) it updates weights using the gradient of the loss function.
- 15) b) The model may overshoot the optimal weight values and fail to converge
- 16) Both (a) and (d)

- 17) a) Decision Tree
- 18) a) The model may suffer from overfitting.
- 19) a) Pruning redundant or less significant rules.
- 20) a) To visualize the performance of a classifier by showing correct and incorrect predictions.
- 21) c) Accuracy
- 22) b) The number of positive instance correctly classified as positive.
- 23) a) $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$.
- 24) b) F1-score
- 25) b) The model has a low number of the false negatives.
- 26) a) Decision Tree
- 27) a) The model may suffer from overfitting.
- 28) a) Pruning redundant or less significant rules.
- 29) a) To visualize the performance of a classifier by showing correct and incorrect predictions.
- 30) c) Accuracy